Shiyun Xiong

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Education

Harbin Institute of Technology **Weihai, China**
Bachelor of Engineering in Software Engineering *Aug. 2021 – Jun. 2025*

- Weighted Average: 90.0/100
- Main course: Linear Algebra(92), Calculus(92), Algorithm Analysis and Design(97), Probability theory and mathematical statistics(90), Introduction to Artificial Intelligence(92), Data Structure(94), Database System(97), Operating Systems(94)

Research Experiences

Adaptive Architecture for Embodied-Sensory Omni-Modal LLMs **Advisor: Prof. Xiangyu Yue**
Multimedia Laboratory, The Chinese University of Hong Kong (MMLab, CUHK) *Jan. 2026 – Present*

- This project aims to develope an adaptive omni-modal LLM architecture for embodied sensory understanding, extending multimodal reasoning from conventional vision-language inputs to heterogeneous sensory modalities such as depth, IMU, pointcloud, thermal, and tactile signals.
- My contributions: Developed the adaptive omni-modal architecture with sensory-specific projection, caption-supervised representation alignment, and modality-aware MoE specialization; curated heterogeneous embodied sensory data; and conducted multi-stage training and comprehensive evaluations.

Autonomous Benchmark Construction Agent for Multimodal Model Evaluation **Advisor: Prof. Xiangyu Yue**
Multimedia Laboratory, The Chinese University of Hong Kong (MMLab, CUHK) *Sep. 2025 – Jan. 2026*

- This project aims to develope the Benchmark Agent, an autonomous agentic system that transforms open-ended evaluation intents into customized multimodal benchmarks through long-horizon planning, tool-augmented reasoning, and self-reflective refinement.
- My contributions: Developed the closed-loop agentic workflow that integrates intent-aware planning, grounded reasoning, constraint-driven execution, and self-verification, enabling scalable generation of reliable benchmarks with controllable evaluation objectives and minimal human supervision.

Self-Evolving Intelligent Agents with Process Rewards in GUI Environments **Jul. 2024 – Feb. 2025**

- This project aims to enhance agent interaction within various visual digital platforms using a proprietary reward model. The goal is to enable agents to adapt based on user interactions and environmental feedback.
- My contributions: Developed the agent-environment interaction pipeline and constructed preference/reward modeling data from multimodal web trajectories. Trained reward models to evaluate agent behaviors and conducted experiments in both static and dynamic visual web environments to validate adaptive decision-making and interaction performance.

Multi-Agent Framework for Web Services with Fuzzy Requirements **Advisor: Prof. Zhiying Tu**
Research Center of Intelligent Computing for Enterprises & Services (ICES, HIT) *Mar 2024 – Jul.2024*

- This project implemented H-Tower, the first LLM-based multi-agent framework simulating multi-turn interactions between humans and websites. It achieves emotion-aware elicitation of complex requirements and mass-produces training data for this task.
- My contributions: Developed the system framework and designed multi-agent collaboration mechanisms to resolve fuzzy user requirements in conversational settings, improving the efficiency and completeness of user need elicitation.

A Benchmark for Universal Instruction Following in Realistic Web Services Navigation **Advisor: Prof. Dianhui Chu**
Research Center of Intelligent Computing for Enterprises & Services (ICES, HIT) *Nov. 2023 – Mar. 2024*

- This project constructed the first Chinese multi-modal benchmark and dataset to evaluate web service navigation tasks and devised an efficient multi-modal framework to evaluate performance on the dataset.
- My contributions: Developed a multimodal web navigation model integrating slot-tree state tracking, instruction parsing, visual grounding, and web execution. Constructed training data with object-detection-based webpage denoising, converted visual UI elements into language representations, and fine-tuned an LLM for improved slot-tree reasoning and task completion.

Publications

Shiyun Xiong, Dongming Wu, Peiwen Sun, Yuang Ai, Bokang Yang, Wencheng Han, Xiao-Hui Li, Xiangyu Yue
"Benchmark Everything Everywhere All at Once." *Under Review*

Bolin Zhang, Dianhui Chu (*Member, IEEE*), **Shiyun Xiong**, Qianjun Gong, and Zhiying Tu (*Member, IEEE*) **"User-Oriented Fuzzy Requirement Clarification for Services Delivery in the Metaverse."** *IEEE Transactions on Services Computing*

Zhiyuan Hu, **Shiyun Xiong**, Yifan Zhang, See-Kiong Ng, Anh Tuan Luu, Bo An, Shuicheng YAN, Bryan Hooi **"Guiding VLM Agents with Process Rewards at Inference Time for GUI Navigation."** *Preprint*

Bolin Zhang, **Shiyun Xiong**, Dianbo Sui, Yunzhe Xu, Zhiying Tu, and Dianhui Chu. 2024. **"RealWeb: A Benchmark for Universal Instruction Following in Realistic Web Services Navigation."** In *2024 IEEE International Conference on Web Services*

Service and Skills

Reviewer: ICLR 2025 Workshop LLM Reason and Plan, ACL ARR 2025 February

Technical Skills and Tools: Python, C++, C, Java, Matlab, Java, Pytorch, SQL, Selenium, Android Studio, etc.

Languages: Chinese Mandarin (naive), English (CET 4: 613, CET 6: 564)